

Heat Transfer in Liquid Metal

Lead–Bismuth Eutectic CFD Analysis of the TALL Facility

SG2224/FSG3224 Applied Computational Fluid Dynamics

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1 Introduction

Lead–bismuth eutectic (LBE) is a binary alloy composed of 44.5 wt% lead and 55.5 wt% bismuth, with a melting point of approximately 398 K (125 °C). It is a prominent candidate for both coolant and spallation target in accelerator-driven systems (ADS), and as a primary coolant for Generation IV Lead-cooled Fast Reactors (LFRs) [OECD2015]. Its high thermal conductivity ($\lambda \approx 11.5 \text{ W m}^{-1} \text{ K}^{-1}$ near 400 K) yields an extremely low thermal Prandtl number ($\text{Pr} \approx 0.02\text{--}0.03$), making molecular heat conduction the dominant heat-transfer mechanism even under turbulent flow conditions.

This characteristic gives rise to a fundamental modelling challenge: in classical forced-convection theory the ratio of the viscous to the thermal boundary layer thickness scales as

$$\frac{\delta_\nu}{\delta_T} \sim \text{Pr}^{1/3}, \quad (1)$$

so for $\text{Pr} \ll 1$ the thermal boundary layer is *much thicker* than the viscous one. This results in difficulties in heat transfer modelling due to the separation of the viscous and the thermal boundary layers. Standard turbulence-based heat-transfer models, calibrated for $\text{Pr} \approx 1$ fluids such as water or air, therefore tend to over-predict the turbulent heat flux and must be used with care.

Investigating the thermal-hydraulic behaviour of LBE and selecting appropriate numerical settings is therefore of great importance. This project investigates two components of the TALL (Thermal-hydraulics ADS Lead–Bismuth Loop) test facility at KTH: the single rod heater and the straight-tube heat exchanger. Both are simplified and modelled in ANSYS Fluent as 2-D axisymmetric, steady-state problems to balance computational efficiency and physical accuracy.

The specific objectives are:

- (i) **Geometry and Material Setup:** Build the 2-D axisymmetric geometries and implement temperature-dependent LBE thermophysical properties via User-Defined Functions (UDFs), ensuring gravity effects and body forces are actively considered.
- (ii) **Mesh Sensitivity:** Conduct a mesh sensitivity study using at least three different mesh sizes, paying special attention to boundary layer refinement to ensure the near-wall y^+ satisfies the chosen turbulence model requirements.
- (iii) **Turbulence Model Comparison:** Test and compare at least two standard RANS turbulence models (e.g., Standard $k\text{--}\epsilon$ and $k\text{--}\omega$ SST) at their default settings using the converged mesh.
- (iv) **Experimental Validation:** First, validate the UDF and mesh strategy on the single rod heater by comparing radial and axial temperature distributions against TALL experimental data and analytical heat balance results. Second, simulate the straight tube heat exchanger to evaluate conjugate heat transfer (fluid-solid-fluid coupled, counter-current flow). After reaching steady state, compare the results with the experimental operating conditions (at $t = 2000 \text{ s}$) and analytical calculations.

2 Theoretical Background

2.1 Governing Equations

In this study, the flow of LBE is described by the incompressible Reynolds-Averaged Navier–Stokes (RANS) equations coupled with the energy equation. Gravity is retained as a body force to account for natural-convection contributions, which are particularly important for the conjugate heat transfer in the straight tube.

Continuity.

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Momentum.

$$\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nabla \cdot \left[\mu_{\text{eff}}(\nabla \mathbf{u} + (\nabla \mathbf{u})^\top) \right] + \rho \mathbf{g}, \quad (3)$$

where $\mu_{\text{eff}} = \mu + \mu_t$ is the effective viscosity (molecular plus turbulent) and $\rho \mathbf{g}$ is the gravitational body force with $g = 9.81 \text{ m s}^{-2}$ directed downward ($-z$). While the Boussinesq approximation is traditionally used to model buoyancy via $\rho \approx \rho_{\text{ref}}[1 - \beta(T - T_{\text{ref}})]$, this study implements the temperature-dependent density directly into the governing equations via a User-Defined Function (UDF): $\rho = 11065 - 1.293 T$.

Energy.

$$\rho c_p (\mathbf{u} \cdot \nabla T) = \nabla \cdot \left[\left(\lambda + \frac{\mu_t c_p}{\text{Pr}_t} \right) \nabla T \right] + \dot{q}, \quad (4)$$

where λ is the molecular thermal conductivity, Pr_t is the turbulent Prandtl number, and $\dot{q}$ is any volumetric heat source (zero in the fluid domain of the present problem). The thermophysical properties ρ , c_p , μ , and λ are dynamically updated according to the UDF.

2.2 Key Dimensionless Parameters

Reynolds number.

$$\text{Re} = \frac{\rho U_{\text{in}} D_h}{\mu}. \quad (5)$$

Prandtl number.

$$\text{Pr} = \frac{\mu c_p}{\lambda} = \frac{\nu}{\alpha}, \quad (6)$$

where $\nu = \mu/\rho$ is the kinematic viscosity and $\alpha = \lambda/(\rho c_p)$ is the thermal diffusivity.

Péclet number.

$$\text{Pe} = \text{Re} \cdot \text{Pr} = \frac{U_{\text{in}} D_h}{\alpha}. \quad (7)$$

The Péclet number is the most relevant parameter governing liquid-metal heat transfer: it compares convective heat transport to molecular conduction. For the single rod heater ($\text{Pr} \approx 0.02$, $\text{Re} \approx 121,000$):

$$\text{Pe} = 0.02 \times 121,000 \approx 2,400. \quad (8)$$

Nusselt number.

$$\text{Nu} = \frac{h D_h}{\lambda}, \quad h = \frac{q_w}{T_w - T_b}. \quad (9)$$

2.3 Turbulent Prandtl Number for Liquid Metals

For water and air, the standard assumption $\text{Pr}_t \approx 0.85$ is adequate. For liquid metals ($\text{Pr} \ll 1$), however, molecular conduction controls heat transport even in turbulent flow, and the turbulent Prandtl number becomes significantly larger than unity. To accurately model this behaviour and assess its impact on the thermal field, two distinct modifications for the turbulent Prandtl number were implemented and compared: the Reynolds modification and the Kays modification [Underhill2021].

The correlation proposed by Reynolds (1975), as detailed in [Underhill2021], computes a single, global value of the turbulent Prandtl number as a function of the bulk Reynolds number (Re) and the bulk Péclet number (Pe):

$$\text{Pr}_t = \left(1 + 100 \text{Pe}^{-1/2} \right) \left(\frac{1}{1 + 120 \text{Re}^{-1/2}} - 0.15 \right). \quad (10)$$

Substituting the operational parameters of the simulation, $\text{Pe} = 2,400$ and $\text{Re} = 121,000$:

$$\text{Pr}_t = \left(1 + \frac{100}{\sqrt{2400}} \right) \left(\frac{1}{1 + 120/\sqrt{121000}} - 0.15 \right) \approx \mathbf{1.8}. \quad (11)$$

A global value of $\text{Pr}_t > 1$ confirms that turbulent momentum transport is more efficient than turbulent heat transport: the eddy thermal diffusivity is smaller than the eddy viscosity, meaning molecular conduction must carry a larger fraction of the total heat flux.

In contrast, the modification proposed by Kays (1994), also described in [Underhill2021], calculates local values of Pr_t throughout the domain as a function of the turbulent Péclet number (Pe_t):

$$\text{Pr}_t = 0.85 + \frac{0.7}{\text{Pe}_t} \quad (12)$$

where the turbulent Péclet number is defined locally based on the kinematic eddy viscosity (ν_t) and the molecular kinematic viscosity (ν):

$$\text{Pe}_t = \text{Pr} \left(\frac{\nu_t}{\nu} \right). \quad (13)$$

This comparison highlights two fundamentally different approaches to modelling LBE heat transfer. The Reynolds correlation imposes a globally evaluated turbulent Prandtl number ($\text{Pr}_t \approx 1.8$), which uniformly suppresses the turbulent heat flux calculation across the entire domain. Conversely, the Kays correlation provides a dynamic field approach: in highly turbulent regions (where $\nu_t \gg \nu$), Pr_t asymptotically approaches the standard 0.85, but in regions with weaker turbulence or near the viscous sublayer, Pr_t increases significantly. Both models were tested in the CFD solver to evaluate the sensitivity of the temperature profiles to constant versus dynamically resolved turbulent thermal diffusivity.

2.4 Analytical Correlations for Liquid-Metal Heat Transfer

Standard correlations such as Dittus–Boelter (valid for $\text{Pr} \geq 0.6$) are inapplicable to LBE. The benchmark correlations used for validation are:

Lyon–Martinelli (uniform wall heat flux, $\text{Pe} > 100$).

$$\text{Nu} = 7.0 + 0.025 \text{Pe}^{0.8}. \quad (14)$$

Sleicher–Rouse.

$$\text{Nu} = 6.3 + 0.0167 \text{Pe}^{0.85} \text{Pr}^{0.08}. \quad (15)$$

At $\text{Pe} = 2,400$, the Lyon–Martinelli correlation gives:

$$\text{Nu}_{\text{LM}} = 7.0 + 0.025 \times (2400)^{0.8} = 7.0 + 0.025 \times 506 \approx 19.6, \quad (16)$$

which provides the target heat transfer coefficient for the single rod validation.

2.5 Turbulence Models

2.5.1 Standard k – ε Model

The Standard k – ε model [Jones1972] solves two transport equations for the turbulent kinetic energy k and its dissipation rate ε :

$$\rho(\mathbf{u} \cdot \nabla)k = \nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \nabla k \right] + P_k - \rho\varepsilon, \quad (17)$$

$$\rho(\mathbf{u} \cdot \nabla)\varepsilon = \nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \nabla \varepsilon \right] + C_{\varepsilon 1} \frac{\varepsilon}{k} P_k - C_{\varepsilon 2} \rho \frac{\varepsilon^2}{k}, \quad (18)$$

with turbulence production $P_k = \mu_t |\nabla \mathbf{u}|^2$ and eddy viscosity $\mu_t = \rho C_\mu k^2 / \varepsilon$. The model constants are:

$$C_\mu = 0.09, \quad C_{\varepsilon 1} = 1.44, \quad C_{\varepsilon 2} = 1.92, \quad \sigma_k = 1.0, \quad \sigma_\varepsilon = 1.3. \quad (19)$$

This model relies on empirical wall functions in the viscous sublayer and defaults to $\text{Pr}_t = 0.85$, making it generally poorly suited for capturing the thick thermal boundary layer of liquid metals.

2.5.2 k – ω SST Model

The SST model [Menter1994] blends the k – ω formulation near walls (superior for adverse-pressure-gradient and low-Re regions) with the k – ε formulation in the free stream:

$$\rho(\mathbf{u} \cdot \nabla)k = \nabla \cdot [(\mu + \sigma_k \mu_t) \nabla k] + \tilde{P}_k - \beta^* \rho \omega k, \quad (20)$$

$$\rho(\mathbf{u} \cdot \nabla)\omega = \nabla \cdot [(\mu + \sigma_\omega \mu_t) \nabla \omega] + \frac{\gamma}{\nu_t} \tilde{P}_k - \beta \rho \omega^2 + 2(1 - F_1) \frac{\rho \sigma_\omega^2}{\omega} (\nabla k \cdot \nabla \omega), \quad (21)$$

where F_1 is a blending function (unity at the wall, zero in the far field). The eddy viscosity includes a stress limiter:

$$\mu_t = \frac{\rho a_1 k}{\max(a_1 \omega, S F_2)}, \quad (22)$$

with S the mean strain rate and F_2 a second blending function. The SST model resolves the viscous sublayer directly (when $y^+ \leq 1$), avoiding standard wall-function assumptions and providing a more robust framework for resolving the separated thermal and viscous boundary layers.

2.5.3 Dimensionless Wall Distance y^+

$$y^+ = \frac{\rho u_\tau \Delta y_1}{\mu}, \quad u_\tau = \sqrt{\frac{\tau_w}{\rho}}, \quad (23)$$

where Δy_1 is the first-cell height and u_τ is the friction velocity. To ensure proper boundary layer resolution:

- Standard $k-\varepsilon$ with standard wall functions requires: $y^+ \approx 30-300$.
- $k-\omega$ SST (low-Re, wall-resolving) requires: $y^+ \simeq 1$.

3 Material Properties of Lead–Bismuth Eutectic

3.1 Temperature-Dependent Correlations

The following correlations, valid at $p = 0.1$ MPa in the range 400 K to 1100 K, are taken from the OECD/NEA handbook [OECD2015]. All temperatures are in Kelvin.

$$\rho(T) = 11065 - 1.293 T \quad [\text{kg m}^{-3}], \pm 0.8\% \quad (24)$$

$$\mu(T) = 4.94 \times 10^{-4} \exp\left(\frac{754.1}{T}\right) \quad [\text{Pa s}], \pm 8\% \quad (25)$$

$$\lambda(T) = 3.284 + 1.617 \times 10^{-2} T - 2.305 \times 10^{-6} T^2 \quad [\text{W m}^{-1} \text{K}^{-1}], \pm 15\% \quad (26)$$

$$c_p(T) = 164.8 - 3.94 \times 10^{-2} T + 1.25 \times 10^{-5} T^2 - 4.56 \times 10^5 T^{-2} \quad [\text{J kg}^{-1} \text{K}^{-1}], \pm 7\% \quad (27)$$

3.2 Evaluated Properties at Inlet (573 K)

Table 1: LBE thermophysical properties at $T_{\text{in}} = 573$ K (300 °C).

Property	Symbol	Correlation	Value	Unit
Density	ρ	Eq. (24)	10,324	kg m^{-3}
Dynamic viscosity	μ	Eq. (25)	1.842×10^{-3}	Pa s
Thermal conductivity	λ	Eq. (26)	11.79	$\text{W m}^{-1} \text{K}^{-1}$
Specific heat	c_p	Eq. (27)	≈ 146	$\text{J kg}^{-1} \text{K}^{-1}$
Prandtl number	Pr	$\mu c_p / \lambda$	0.023	—

3.3 UDF Implementation in ANSYS Fluent

All four temperature-dependent thermophysical properties (ρ , μ , λ , and c_p) are implemented using `DEFINE_PROPERTY` macros, which are compiled as C-language User-Defined Functions (UDFs) within the ANSYS Fluent solver. The complete source code for these property correlations is provided in Appendix A.

Additionally, to correctly evaluate the Boussinesq body force driven by the temperature-dependent density, gravity ($g_z = -9.81 \text{ m s}^{-2}$) is actively enabled in the solver. The operating density is explicitly set to the inlet condition, $\rho(T_{\text{in}}) = 10324 \text{ kg m}^{-3}$, to serve as the baseline reference for buoyancy calculations.

4 Single Rod Heater

4.1 Geometry and Boundary Conditions

The single rod heater section of the TALL facility is modelled as a 2-D axisymmetric annular channel. The computational domain represents a fluid channel bounded by the active heater rod on the inside and an insulated housing on the outside. To ensure the flow is fully developed before entering the heated region, accurately reflecting the real facility, an unheated inlet section of 100 mm is added upstream of the active length.

The LBE fluid enters the bottom of the test section and flows upward. To accurately capture the hydrostatic head of the full TALL loop ($H_{\text{TALL}} = 6.8$ m, with the heater top at $h_{\text{top}} = 2.1$ m), specific gauge pressures are applied at the

boundaries based on the fluid column height above the heater. Thermal boundary conditions consist of a constant heat flux applied to the inner wall to simulate the heater power, while the outer housing wall and the unheated entrance region are assumed to be perfectly insulated (adiabatic).

A comprehensive summary of the geometric dimensions, operating boundary conditions, and the resulting dimensionless flow parameters (evaluated at the inlet temperature) is provided in Table 2 and shown in Figure 1.

Table 2: Geometric, operational, and flow parameters for the single rod heater model.

Parameter	Symbol	Value	Unit
<i>Geometry</i>			
Unheated inlet length	L_{inlet}	100	mm
Active heated length	L	870	mm
Inner heater radius	r_i	4.1	mm
Outer housing radius	r_o	9.5	mm
Annular flow gap	—	5.4	mm
Hydraulic diameter	D_h	10.8	mm
Flow cross-section	A	2.31×10^{-4}	m ²
<i>Boundary Conditions</i>			
Inlet velocity	U_{in}	2.0	m s ⁻¹
Inlet temperature	T_{in}	573	K (300 °C)
Inlet gauge pressure	p_{inlet}	$\approx 564,121$	Pa
Outlet gauge pressure	p_{outlet}	$\approx 476,014$	Pa
Total heater power	Q	21	kW
Inner wall heat flux	q_w	≈ 937	kW m ⁻²
Outer wall	—	Adiabatic	—
<i>Dimensionless Flow Parameters</i>			
Prandtl number	Pr	0.023	—
Reynolds number	Re	$\approx 121,000$	—
Péclet number	Pe	$\approx 2,400$	—

4.2 Mesh Generation and Sensitivity Study

4.2.1 Near-Wall Sizing: y^+ Estimate

The target first-cell height for $y^+ = 1$ is estimated from the friction velocity u_τ computed via the Haaland equation for the friction factor f :

$$\frac{1}{\sqrt{f}} = -1.8 \log_{10} \left[\left(\frac{k_s/D_h}{3.7} \right)^{1.11} + \frac{6.9}{\text{Re}} \right], \quad (28)$$

where $k_s = 0.005$ mm is a common roughness height for stainless steel. With $\text{Re} = 121,000$ and $D_h = 10.8$ mm:

$$f = 19.4 \times 10^{-3}, \quad C_f = \frac{f}{4} = 4.8 \times 10^{-3}. \quad (29)$$

The wall shear stress and friction velocity:

$$\tau_w = C_f \cdot \frac{1}{2} \rho U^2 = 4.8 \times 10^{-3} \times \frac{1}{2} \times 10,324 \times 2^2 \approx 100 \text{ Pa}, \quad (30)$$

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}} = \sqrt{\frac{100}{10,324}} \approx 0.098 \text{ m s}^{-1}. \quad (31)$$



Figure/geometry_singleRod.png

Figure 1: Geometry and section view of the single rod heater.

The viscous length scale and required first-cell height for $y^+ = 1$:

$$\ell^* = \frac{\mu}{\rho u_\tau} = \frac{1.842 \times 10^{-3}}{10,324 \times 0.098} \approx 1.8 \mu\text{m}. \quad (32)$$

Therefore, $\Delta y_1 = y^+ \cdot \ell^* = 1 \times 1.8 \mu\text{m} \approx 0.002 \text{ mm}$.

4.2.2 Mesh Refinement Levels and Error Analysis

To ensure the numerical solution is independent of the spatial discretization, four distinct mesh refinement levels were generated. The boundary layer resolution systematically increases from a coarse grid suited for standard wall functions ($y^+ > 30$) to highly refined grids capable of explicitly resolving the viscous sublayer ($y^+ \approx 1$), which is crucial for the low-Re SST $k-\omega$ formulation. The properties of the four meshes are summarized in Table 3 and the finest mesh grid is shown in Figures 2a and 2b.

Table 3: Mesh configurations for the single rod heater sensitivity study.

Mesh	Δy_1 (μm)	Layers	Growth rate	Max Size (mm)	Total cells
Mesh 0	150	6	1.20	0.5	34,920
Mesh 1	40	15	1.15	0.5	64,020
Mesh 2	4	30	1.15	0.5	124,160
Mesh 3	2	35	1.15	0.2	388,000

Rather than using a theoretical extrapolation method, grid independence is evaluated through a direct statistical error analysis. The finest grid (Mesh 3, with 388,000 cells) is designated as the reference solution. Axial and radial temperature profiles extracted from the coarser meshes (Meshes 0, 1, and 2) are linearly interpolated onto the spatial coordinates of the reference mesh.

The deviation of each coarse mesh from the reference solution is then quantified using four statistical metrics:

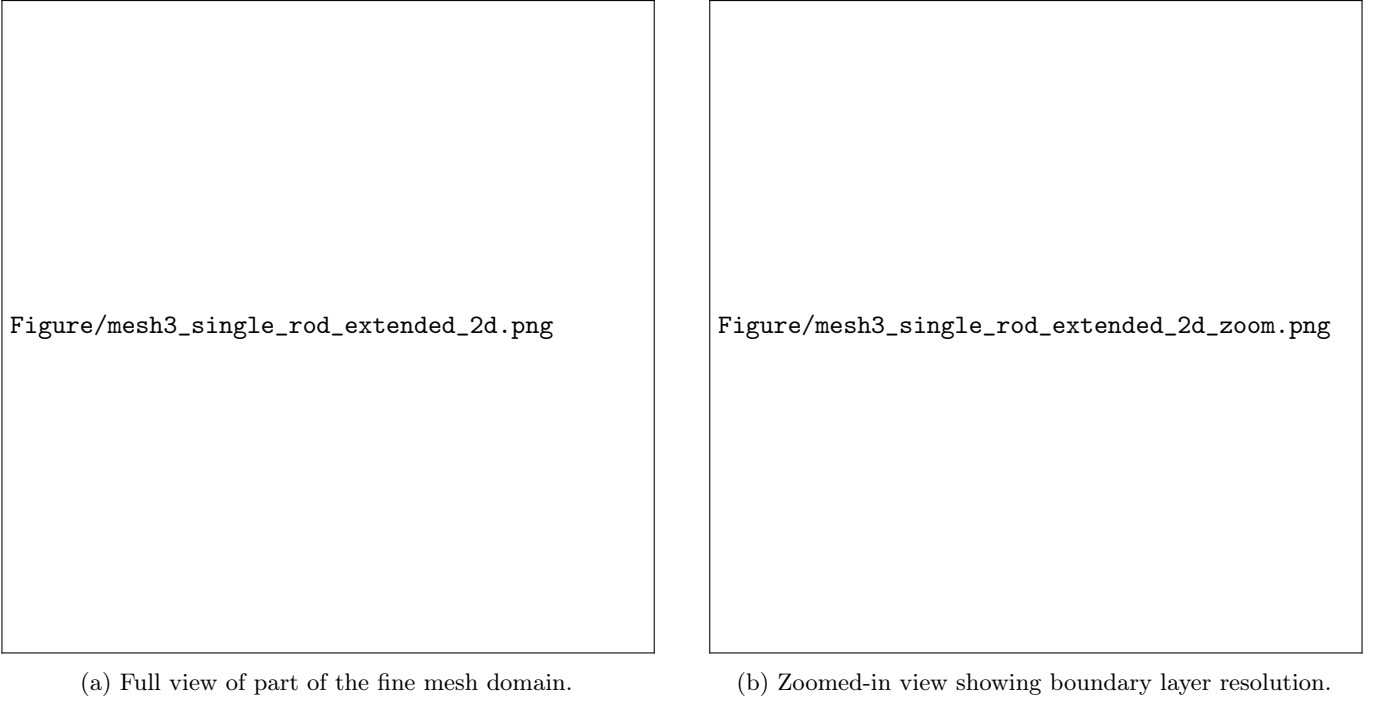


Figure 2: Overview of the finest mesh configuration (388,000 cells) utilized for the grid sensitivity study, designed to maintain a $y^+ \approx 1$ at the heated wall.

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Maximum Absolute Error ($\max |T_{\text{mesh}} - T_{\text{ref}}|$)
- Mean Relative Error (MRE, expressed as a percentage)

By plotting these error metrics as a function of the total cell count, the asymptotic convergence of the temperature field can be verified, allowing for the selection of an optimal mesh that balances computational cost with numerical accuracy.

4.3 ANSYS Fluent Solver Configuration

Table 4: ANSYS Fluent solver settings for the single rod heater.

Setting	Value
Solver	Pressure-based, steady-state, axisymmetric
Pressure-velocity coupling	Coupled
Pressure discretisation	Second-order
Momentum discretisation	Second-order upwind
Energy discretisation	Second-order upwind
Turbulence ($k, \varepsilon/\omega$)	First-order upwind
Gradient scheme	Least-squares cell-based
Gravity	$g_z = -9.81 \text{ m s}^{-2}$
Operating density	$\rho(T_{\text{in}}) = 10.324 \text{ kg m}^{-3}$
Operating pressure	101 325 Pa

4.3.1 Turbulence Modelling Strategy

To evaluate the impact of different turbulence closures on the liquid metal heat transfer, three distinct modelling setups were simulated:

1. **Standard $k-\varepsilon$ Model:** Configured with Enhanced Wall Treatment (EWT) and thermal effects activated to

better resolve the near-wall thermal boundary layer. The choice of using EWT also allowed us to use the same $y^+ \approx 1$ mesh as for the other methods.

2. **$k-\omega$ SST Model:** Used as the baseline for low-Re boundary layer resolution.
3. **Kays Correlation (UDF):** A modified setup where the default constant turbulent Prandtl number ($Pr_t \approx 1.8$) was replaced by a UDF implementing the Kays dynamic correlation for the energy Prandtl number, while maintaining the default $Pr_t = 0.85$ for the wall Prandtl number.

4.3.2 Solution Strategy and Convergence

Rather than enforcing strict residual thresholds, convergence was evaluated by monitoring the asymptotic stabilization of the residual plots and key integral variables (e.g., area-averaged outlet temperature and heat transfer balance).

To ensure numerical stability on the highly refined grids (Mesh 2 and Mesh 3), a phased pseudo-transient time-stepping strategy with strict under-relaxation was employed:

- **Phase 1 (Iterations 1–800):** A highly constrained setup using a time step of 0.01 s. The under-relaxation factors (URFs) were set to: Pressure 0.1, Momentum 0.3, Density 1.0, Body Forces 0.5, Turbulent Kinetic Energy (k) 0.5, Specific Dissipation Rate (ω/ε) 0.5, Turbulent Viscosity 1.0, and Energy 0.5.
- **Phase 2 (Iterations 801–1800):** The time step was increased to 0.1 s for the next 1000 iterations while maintaining the initial URFs.
- **Phase 3 (Until Convergence):** To accelerate final convergence, the time step was increased to 0.5 s. The URFs for the turbulence and energy equations were relaxed to: Turbulent Kinetic Energy 0.75, Specific Dissipation Rate 0.75, and Energy 0.7.

For the coarser grids (Mesh 0 and Mesh 1), the high spatial numerical stability allowed the simulations to be run directly using the final Phase 3 configuration (larger time step and relaxed URFs) from initialization until convergence. Figure 3 shows the residual plot from a simulation using Mesh 2 and the $k-\omega$ SST model with the Kays correlation. This plot serves as a representative example of the convergence criteria met throughout the entire study.

4.4 Analytical Heat Balance

A global energy balance between the inlet and the outlet is calculated to verify the overall energy conservation of the numerical model:

$$Q = \dot{m} c_p (T_{\text{out}} - T_{\text{in}}). \quad (33)$$

Using the corrected flow cross-section $A = 2.31 \times 10^{-4} \text{ m}^2$, the mass flow rate $\dot{m} = \rho U_{\text{in}} A$ evaluates to:

$$\dot{m} = 10,324 \times 2.0 \times 2.31 \times 10^{-4} \approx 4.77 \text{ kg s}^{-1}. \quad (34)$$

The expected bulk temperature rise across the active heated length is:

$$\Delta T = \frac{Q}{\dot{m} c_p} = \frac{21,000}{4.77 \times 146} \approx 30.15 \text{ K}, \quad (35)$$

yielding an expected analytical outlet bulk temperature of $T_{\text{out}} \approx 330.15^\circ\text{C}$. This analytical value serves as a primary macro-scale validation metric for the converged CFD results.

4.5 Results and Validation

4.5.1 Mesh Sensitivity Analysis and Error Computation

Grid independence was evaluated via a direct statistical error analysis, utilizing a custom post-processing script. The finest grid (Mesh 3, 388,000 cells) was designated as the baseline "ground truth" reference solution. Because the spatial nodes of the coarser meshes (Mesh 0, 1, and 2) do not perfectly align with the nodes of the finest mesh, a spatial mapping step was required before computing the error.

Let $T_{\text{ref}}(x_i)$ be the temperature profile of the reference mesh evaluated at N discrete spatial coordinates x_i , and let $T_m(x)$ be the temperature profile of a coarser mesh m . To perform a node-to-node comparison, the coarse mesh temperatures were linearly interpolated (with linear extrapolation at the boundaries) onto the reference grid coordinates to yield $T_{m,\text{interp}}(x_i)$.

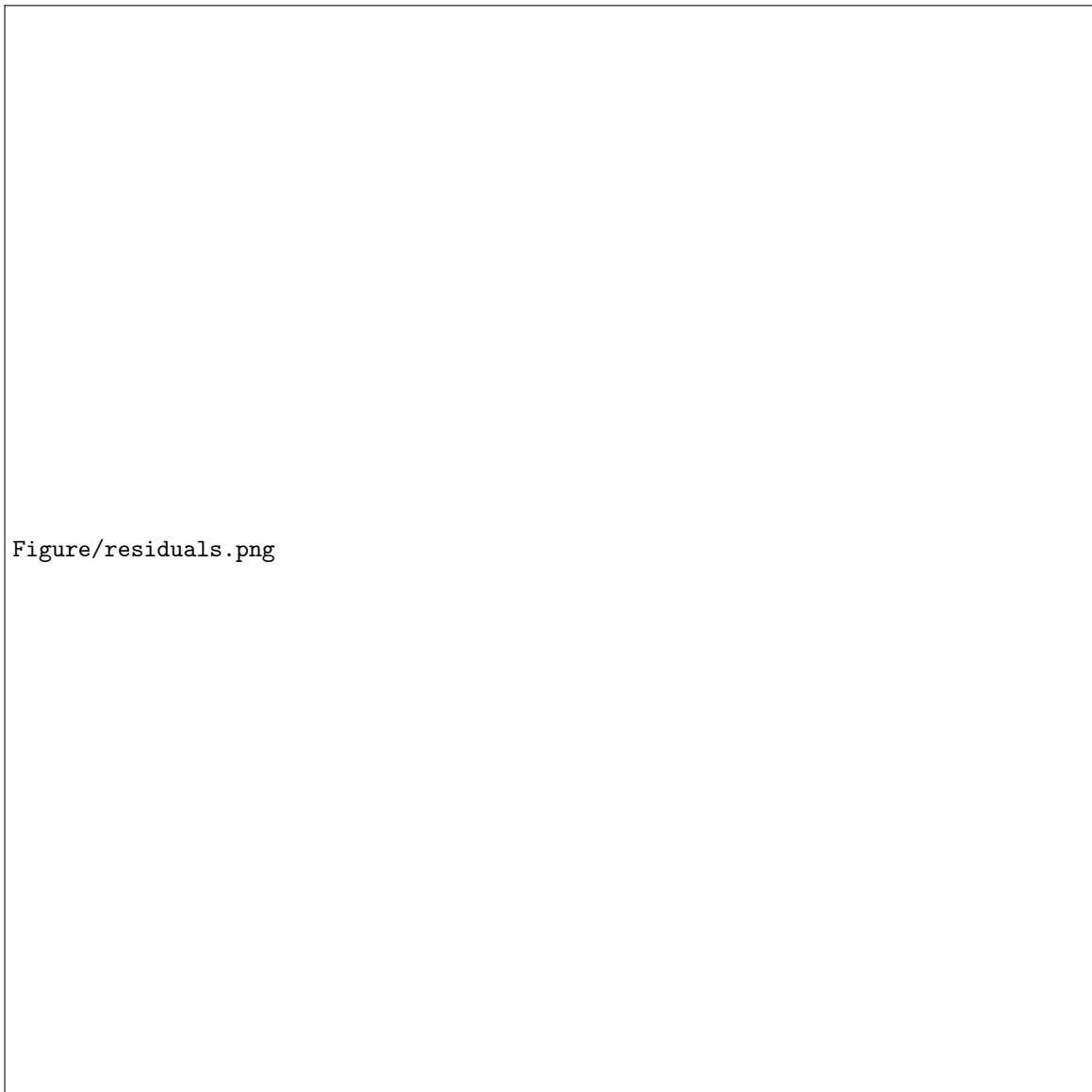


Figure 3: Scaled residuals monitored during the steady-state simulation of Mesh 2.

The local absolute error at each discrete point i is defined as:

$$e_i = |T_{m,\text{interp}}(x_i) - T_{\text{ref}}(x_i)|. \quad (36)$$

The global deviation of each coarse mesh from the reference solution was then quantified using four statistical metrics:

$$\text{Mean Absolute Error (MAE)} = \frac{1}{N} \sum_{i=1}^N e_i \quad (37)$$

$$\text{Root Mean Square Error (RMSE)} = \sqrt{\frac{1}{N} \sum_{i=1}^N e_i^2} \quad (38)$$

$$\text{Maximum Absolute Error} = \max_i(e_i) \quad (39)$$

$$\text{Mean Relative Error (MRE \%)} = \frac{\text{MAE}}{\frac{1}{N} \sum_{i=1}^N |T_{\text{ref}}(x_i)|} \times 100 \quad (40)$$

By plotting these error metrics as a function of the total cell count (Figures 4 and 5), the numerical error exhibits clear convergence as the grid is refined. Even for the coarsest grid, the Mean Relative Error (MRE) remains remarkably low ($< 0.2\%$ for radial profiles and $< 0.15\%$ for axial profiles). However, the absolute error metrics (MAE, RMSE, and Max Error) show a significant drop between Mesh 1 (64,020 cells) and Mesh 2 (124,160 cells). Because Mesh 2 successfully reduces the maximum absolute error to below 1 K while achieving the sufficiently low y^+ values ($y^+ \approx 1$) required by the $k-\omega$ SST turbulence model, it provides an optimal balance between computational cost and physical accuracy for the remainder of the study.

4.5.2 Validation against TALL Experimental Data

While the CFD model captures the general phenomenological trends of the LBE flow, a critical analysis of Figures 6 and 7 reveals significant quantitative discrepancies between the numerical predictions and the TALL experimental data. Rather than indicating a perfect match, these deviations highlight the inherent limitations of both the idealized geometric model and standard RANS formulations for low-Prandtl fluids.

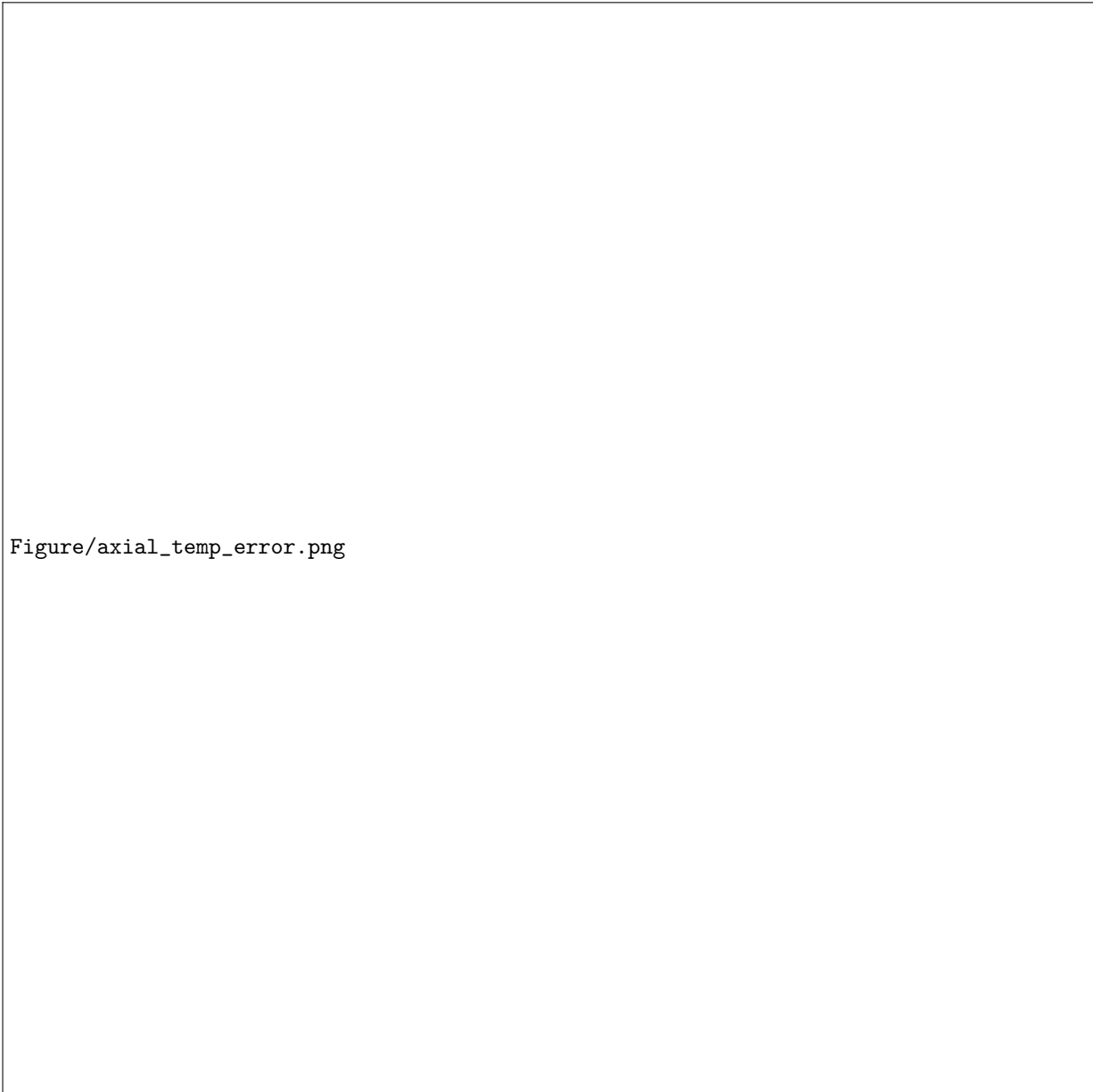
The observed discrepancies can be primarily attributed to two major factors:

1. Inlet Conditions and Unmodeled Heat Losses As observed in the axial temperature profiles (Figure 7), the experimental temperatures exhibit both a different initial offset and a noticeably slower rate of rise compared to the CFD predictions. First, an analysis of the experimental data points (extracted via WebPlotDigitizer) reveals that the actual inlet temperature in the physical facility was slightly higher than the nominal 300°C (573.15 K) prescribed by the simulation boundary conditions.

Despite starting at a higher initial temperature, the experimental fluid temperatures rise at a much slower rate and ultimately fall well below the CFD predictions by the end of the heated section. This flattened gradient directly highlights the impact of unmodelled parasitic heat losses. The numerical setup assumes a perfectly adiabatic outer housing, mathematically forcing the entirety of the 21 kW heat load into the LBE fluid. In the actual facility, thermal energy inevitably escapes to the ambient environment, reducing the net power absorbed by the fluid. This combined effect—a slightly offset actual inlet temperature coupled with non-adiabatic heat losses—explains why the experimental and simulated axial curves have fundamentally different slopes.

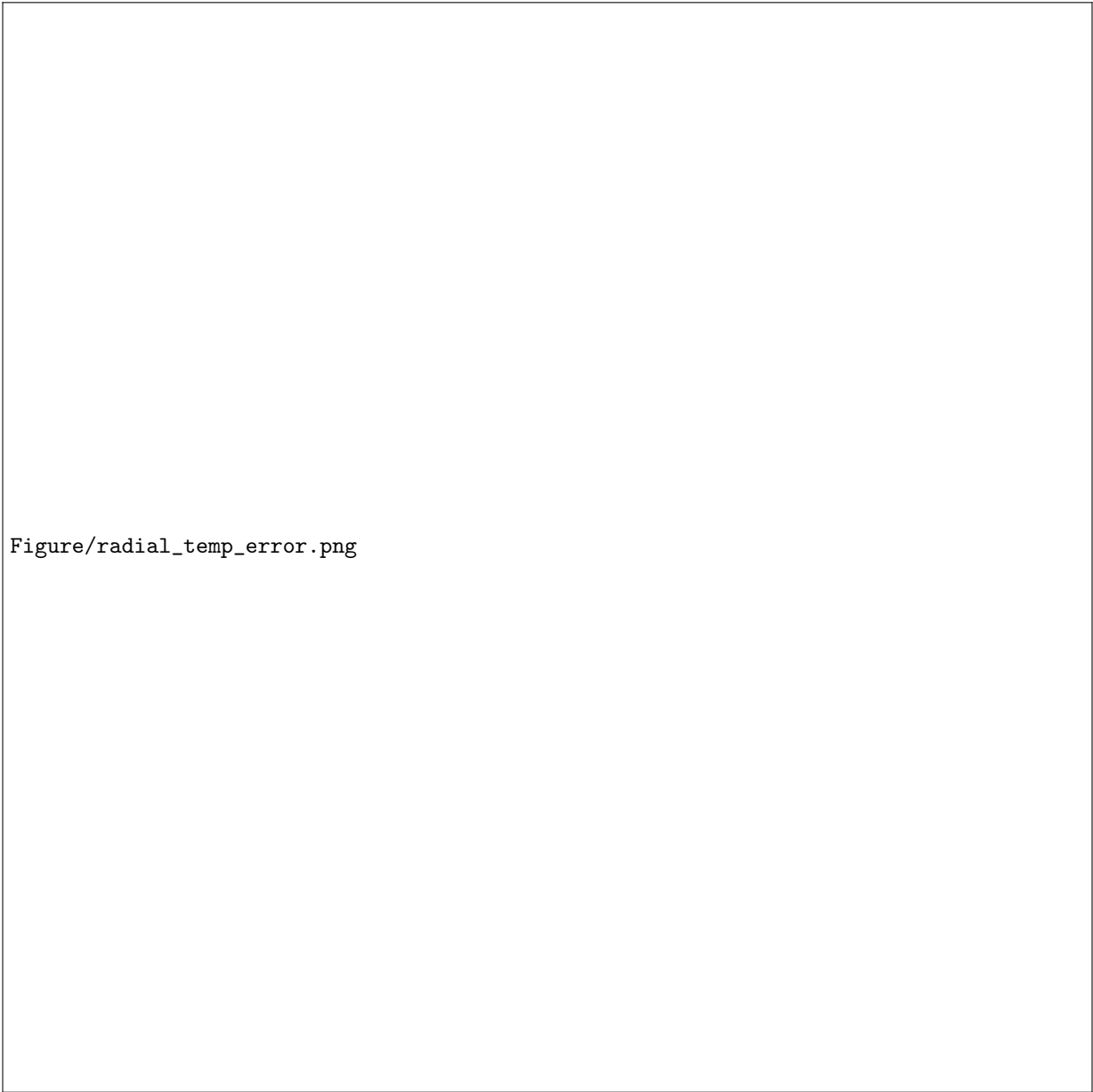
2. Limitations in Modeling Liquid Metal Turbulence A fundamental limitation of the numerical model is evident in the radial temperature distributions (Figure 6). While the experimental measurements display the highly uniform, flat profiles characteristic of liquid metals, the CFD results predict pronounced temperature gradients near the heated wall. This discrepancy suggests that the use of a common RANS model paired with a globally constant turbulent Prandtl number ($\text{Pr}_t \approx 1.8$) insufficiently captures the true effective radial thermal mixing of the LBE.

Standard turbulence formulations inherently struggle with the decoupled nature of the thermal and viscous boundary layers in low-Prandtl fluids. By enforcing a high, uniform Pr_t throughout the domain, the model artificially suppresses the turbulent thermal diffusivity in the bulk flow. Consequently, heat is not transported radially away from the wall efficiently enough, resulting in a localized temperature over-prediction near the heater and an under-prediction in the channel core. This constrained radial diffusion directly translates to the axial profiles shown in Figure 7, where the



Figure/axial_temp_error.png

Figure 4: Statistical error metrics for axial temperature profiles as a function of total cell count.



Figure/radial_temp_error.png

Figure 5: Statistical error metrics for radial temperature profiles as a function of total cell count.

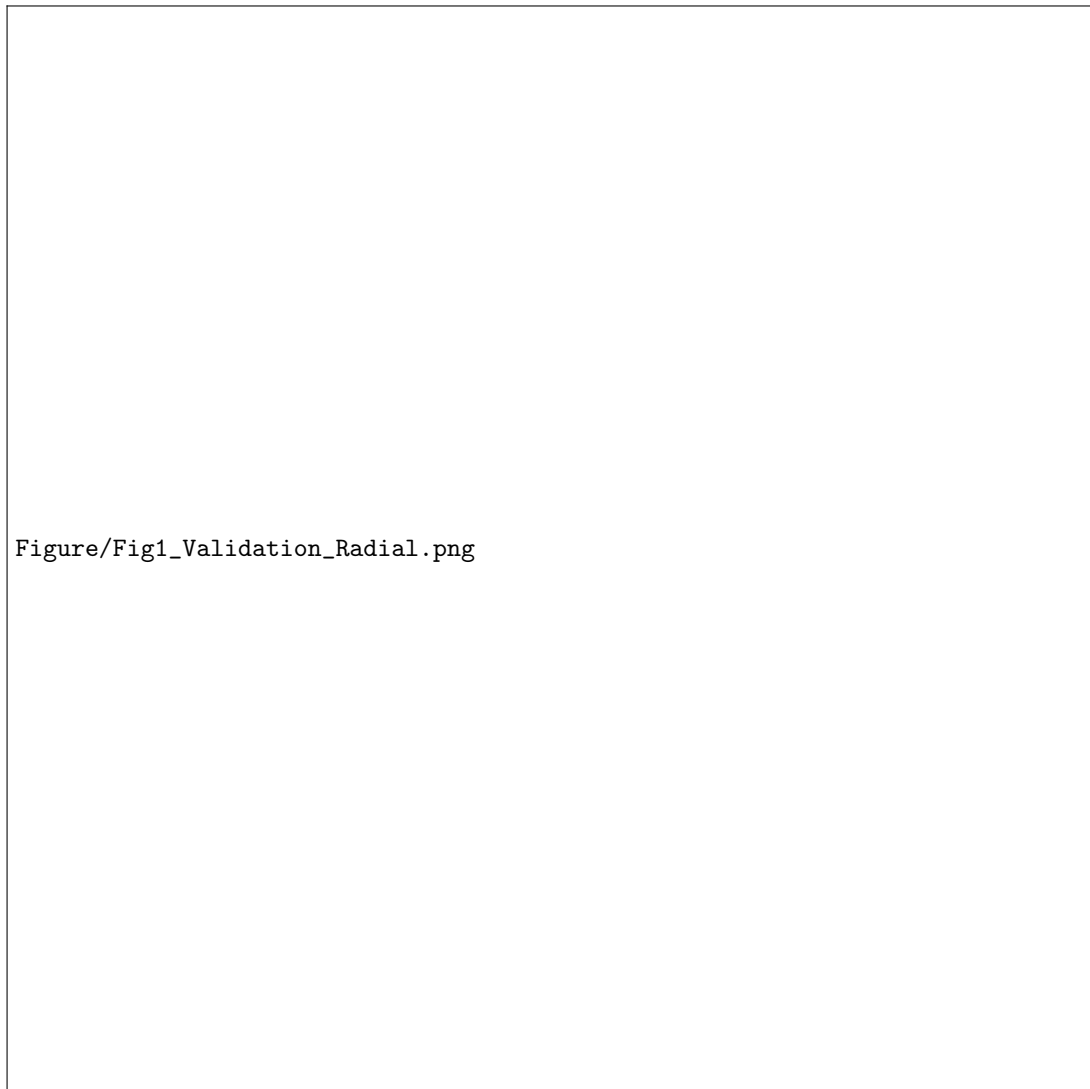


Figure 6: Comparison of simulated (Standard SST) and experimental radial temperature profiles at various axial elevations.

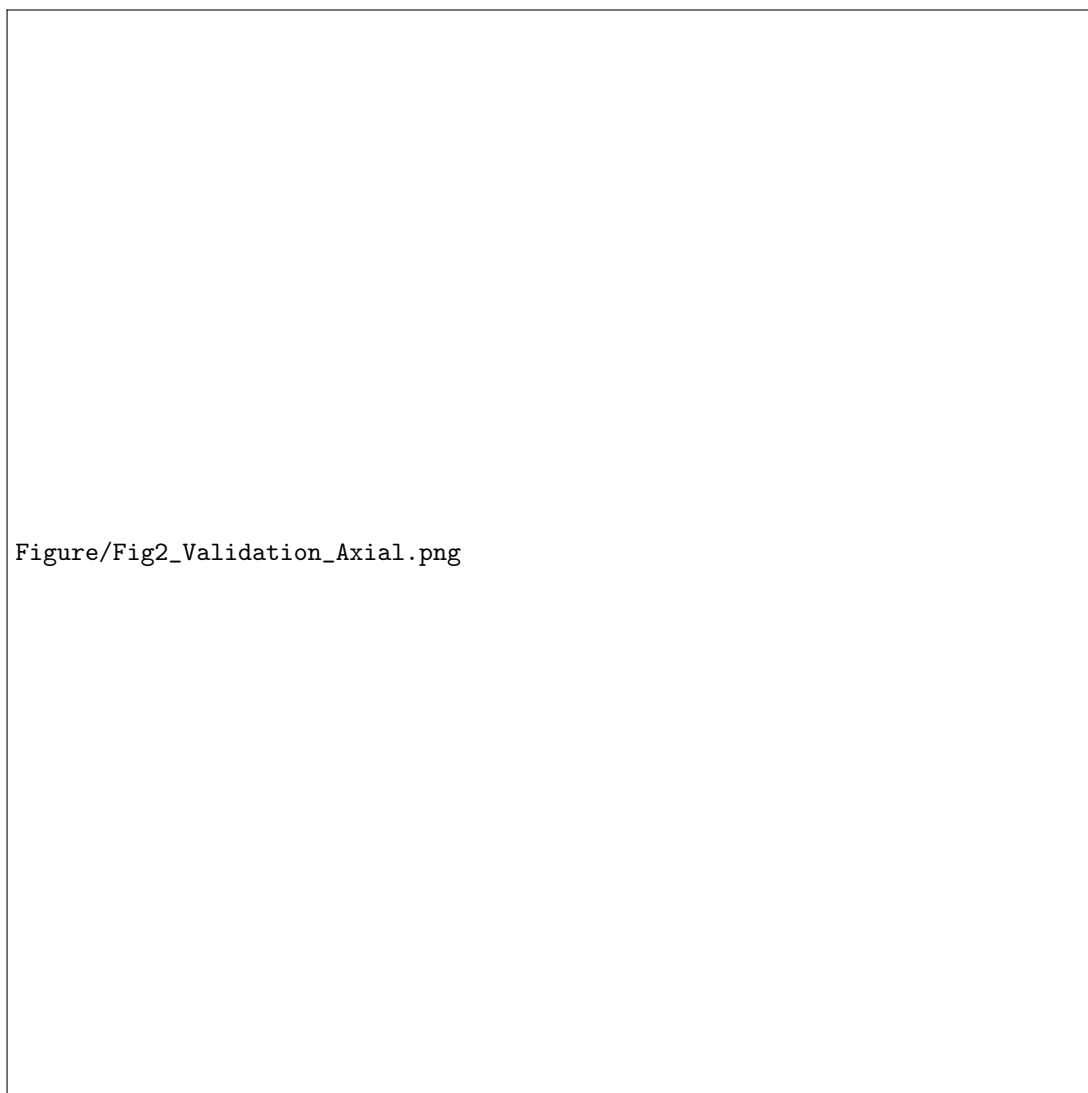


Figure 7: Comparison of simulated (Standard SST) and experimental axial temperature profiles at various radial positions.

simulated curves for different radii diverge widely, in stark contrast to the tightly grouped experimental data points that reflect a much more thoroughly mixed fluid.

4.5.3 Turbulence Model Comparison: $k-\omega$ SST vs. $k-\varepsilon$

Figure/Fig5_Comparison_Radial_Std_vs_kEps.png

Figure/Fig6_Comparison_Axial_Std_vs_kEps.png

Figure 8: Effect of the RANS turbulence model on radial temperature profiles (Standard SST vs. $k-\varepsilon$).

Figure 9: Effect of the RANS turbulence model on axial temperature profiles (Standard SST vs. $k-\varepsilon$).

The impact of the chosen RANS closure model is assessed in Figures 8 and 9, where the Standard $k-\omega$ SST model (solid lines) is compared against the $k-\varepsilon$ model with Enhanced Wall Treatment (dashed lines). As anticipated by theoretical limitations, the $k-\varepsilon$ model systematically predicts even higher temperatures in the near-wall region. The $k-\varepsilon$ formulation, worse at modeling the near-wall regions, severely misrepresents the thick thermal boundary layer characteristic of LBE.

While the wall-resolving capability of the $k-\omega$ SST model provides a comparatively more rigorous numerical treatment of the viscous sublayer, it is crucial to emphasize that both models ultimately fail to accurately capture the real temperature distributions observed in the experiment. Because both $k-\varepsilon$ and $k-\omega$ SST are built upon eddy-viscosity assumptions for standard fluids and rely on a globally elevated turbulent Prandtl number ($Pr_t \approx 1.8$), they both fundamentally under-predict the actual radial thermal mixing of the liquid metal. Therefore, while $k-\omega$ SST represents a slight mathematical improvement over $k-\varepsilon$ near the wall, neither standard RANS formulation is physically equipped to replicate the highly uniform thermal field of the real LBE flow.

4.5.4 Turbulent Prandtl Number Modelling: Reynolds vs. Kays

Finally, the sensitivity of the thermal field to the turbulent Prandtl number formulation is evaluated in Figures 10 and 11. The baseline model utilizes the global Reynolds modification ($Pr_t \approx 1.8$), while the alternative setup applies the Kays correlation to calculate a dynamic, local Pr_t based on the turbulent Péclet number.

The results demonstrate that the Kays formulation yields noticeably lower temperatures near the heated wall and slightly higher temperatures in the bulk fluid. By dynamically adjusting Pr_t , the Kays model predicts a higher and more physically realistic effective turbulent thermal diffusivity in the boundary layer transition region than the fixed Reynolds approach. This allows for enhanced radial thermal mixing, directly mitigating the artificial heat accumulation at the wall seen in the baseline model.

To definitively summarize these findings, Figure 12 presents a comprehensive overlay of all tested numerical models against the experimental data for the axial temperature profile taken at a radial distance of 5.1 mm from the axis of symmetry. This master comparison clearly illustrates the hierarchy of model accuracy. The Standard $k-\varepsilon$ model exhibits the most severe over-prediction of the fluid temperature. The baseline $k-\omega$ SST model offers a slight improvement due to its sublayer-resolving capabilities, but still significantly overestimates the heat accumulation. The Kays modification,

Figure/Fig3_Comparison_Radial_Std_vs_Kays.png

Figure/Fig4_Comparison_Axial_Std_vs_Kays.png

Figure 10: Comparison of radial temperature profiles using a global Pr_t (Standard/Reynolds) versus a local Pr_t (Kays UDF).

Figure 11: Comparison of axial temperature profiles using a global Pr_t (Standard/Reynolds) versus a local Pr_t (Kays UDF).

however, demonstrates the best overall performance, shifting the temperature curve substantially downward to closely trace the experimental data points. Consequently, utilizing a locally resolved Pr_t provides the most accurate and physically realistic representation of the LBE heat transfer among the tested configurations.

5 Straight Tube Heat Exchanger

5.1 Geometry and Boundary Conditions

The straight-tube heat exchanger is modelled as a conjugate heat transfer (CHT) problem consisting of three coupled computational domains: the inner LBE flow, the solid 316L stainless-steel tube wall, and the outer glycerol flow in a counter-current arrangement. To ensure the ANSYS Fluent solver automatically recognizes the fluid-solid interfaces and accurately resolves the thermal coupling across them, a conformal mesh topology (Shared Topology) is enforced prior to volume meshing.

A comprehensive summary of the geometric dimensions and operational boundary conditions is provided in Table 5. The outer housing diameter is derived from the specified annulus hydraulic diameter ($D_{h, gly} = D_{housing} - D_{out}$).

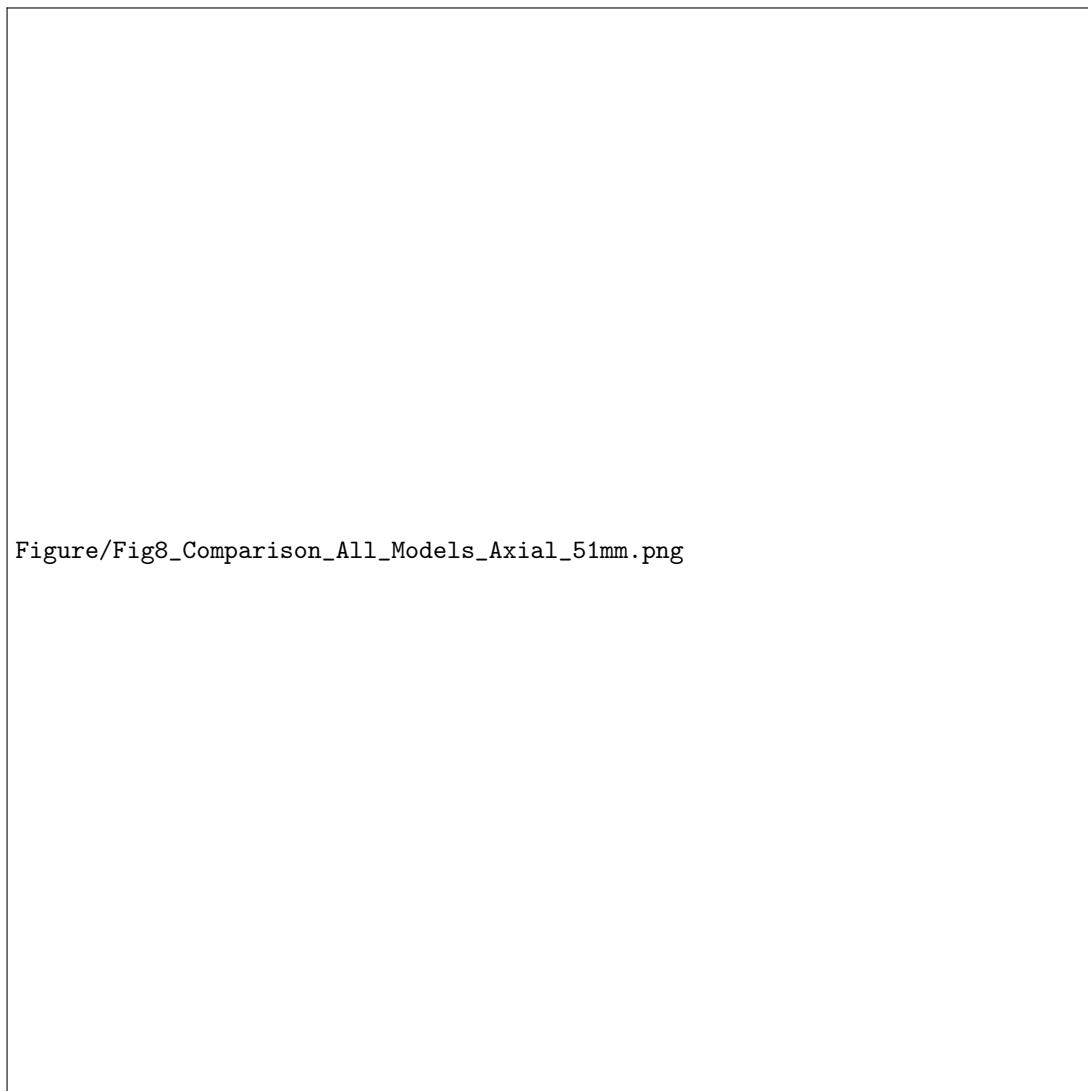


Figure 12: Comprehensive comparison of all tested numerical models (Standard $k-\omega$ SST, $k-\epsilon$, and Kays Modification) against experimental data for the axial temperature profile at a distance of 51 mm from the central axis.

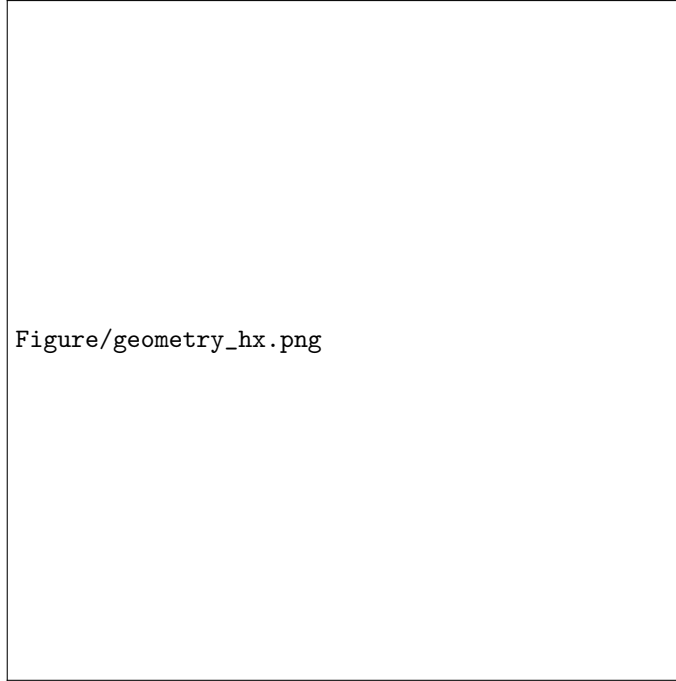


Figure 13: Geometry and section view of the heat exchanger.

Table 5: Geometric and operational parameters for the heat exchanger model.

Parameter	Symbol	Value	Unit
<i>Geometry</i>			
Active length	L	1000	mm
Inner tube inner diameter	D_{in}	5.0	mm
Inner tube outer diameter	D_{out}	5.5	mm
Outer housing inner diameter	D_{housing}	11.5	mm
Glycerol hydraulic diameter	$D_{h,\text{gly}}$	6.0	mm
<i>Boundary Conditions</i>			
LBE inlet velocity	U_{LBE}	0.3	m s^{-1}
LBE inlet temperature	$T_{\text{LBE,in}}$	573	K (300 °C)
LBE outlet gauge pressure	$p_{\text{LBE,out}}$	131,660	Pa
Glycerol inlet velocity	U_{gly}	2.6	m s^{-1}
Glycerol inlet temperature	$T_{\text{gly,in}}$	423.15	K (150 °C)
Glycerol outlet gauge pressure	$p_{\text{gly,out}}$	3,708	Pa
Outer wall	Adiabatic, no-slip		—

5.2 Flow Parameters

Inner LBE Flow Based on the LBE properties evaluated at the inlet temperature of 300 °C ($\rho = 10.324 \text{ kg m}^{-3}$, $\mu = 1.842 \times 10^{-3} \text{ Pa s}$, $\text{Pr} = 0.023$) and an inlet velocity $U_{\text{LBE}} = 0.3 \text{ m s}^{-1}$ within the $D_{\text{in}} = 5 \text{ mm}$ tube, the dimensionless flow parameters are evaluated as:

$$\text{Re}_{\text{LBE}} = \frac{\rho U_{\text{LBE}} D_{\text{in}}}{\mu} = \frac{10,324 \times 0.3 \times 0.005}{1.842 \times 10^{-3}} \approx 8,407, \quad (41)$$

$$\text{Pe}_{\text{LBE}} = \text{Re}_{\text{LBE}} \cdot \text{Pr} = 8,407 \times 0.023 \approx 193. \quad (42)$$

Because $\text{Re}_{\text{LBE}} > 4,000$, the inner LBE flow operates in the fully turbulent regime.

Outer Glycerol Flow For the annular glycerol stream, the fluid properties must be evaluated at the operating inlet temperature of 150 °C (423.15 K), where the fluid thins significantly compared to standard room-temperature conditions. Using the high-temperature properties ($\rho \approx 1175 \text{ kg m}^{-3}$, $\mu \approx 4.5 \times 10^{-3} \text{ Pa s}$, $\text{Pr} \approx 45$) and an inlet velocity $U_{\text{gly}} = 2.6 \text{ m s}^{-1}$ within the hydraulic diameter $D_{h,\text{gly}} = 6 \text{ mm}$:

$$\text{Re}_{\text{gly}} = \frac{\rho U_{\text{gly}} D_{h,\text{gly}}}{\mu} = \frac{1175 \times 2.6 \times 0.006}{4.5 \times 10^{-3}} \approx 4,073. \quad (43)$$

Because the viscosity drops drastically at this elevated temperature, the Reynolds number increases significantly ($\text{Re}_{\text{gly}} > 4,000$), placing the outer glycerol stream into the turbulent flow regime, which greatly enhances the convective heat transfer across the outer tube wall.

5.3 Mesh Generation

The computational grid for the straight-tube heat exchanger was generated using a fully structured quadrilateral mesh, mirroring the rigorous methodology applied to the single rod heater. The global Physics preference was set to CFD (Fluent), utilizing a Linear element order and Quad (2D) element shapes. Global mesh sizing parameters were defined with a minimum element size of 0.001 mm, a maximum size of 0.1 mm, and a bulk growth rate of 1.15, with Capture Curvature enabled to preserve geometric fidelity.

To accurately capture the steep temperature and velocity gradients at the fluid-solid interfaces, inflation layers were carefully applied to both the LBE and glycerol domains. The overall mesh structure, including a full domain view and a detailed cross-section of the conjugate heat transfer boundary layers, is depicted in Figure 14.

5.3.1 Boundary Layer and Solid Resolution

Inflation on the LBE side (inner wall). To explicitly resolve the viscous sublayer and achieve a target $y^+ \leq 1$ for the SST $k-\omega$ turbulence model, inflation was scoped to the LBE fluid body at the inner tube wall:

- First layer height (Δy_1): 0.003 mm
- Number of layers: 28
- Growth rate: 1.12

Based on the LBE Reynolds number ($\text{Re} \approx 8,407$), the viscous length scale is analytically estimated as $\ell^* \approx 0.0092 \text{ mm}$. Because the chosen first cell height ($\Delta y_1 = 0.003 \text{ mm}$) is significantly smaller than ℓ^* , a sublayer resolution of $y^+ \approx 0.3$ is guaranteed across the entire heated length.

Inflation on the glycerol side (outer wall and interface). At the operating temperature of 150 °C, the outer glycerol stream is in a transitional/turbulent regime ($\text{Re} \approx 4,073$) with a moderately high Prandtl number ($\text{Pr} \approx 45$). For fluids with $\text{Pr} \gg 1$, the thermal boundary layer is exceptionally thin and lies deeply embedded within the viscous sublayer. Consequently, explicitly resolving the momentum sublayer ($y^+ \leq 1$) is absolutely critical for accurate heat transfer evaluation. Inflation was scoped to the glycerol body with the following parameters:

- Fluid-solid interface: $\Delta y_1 = 0.01 \text{ mm}$, 15 layers, 1.15 growth rate.
- Outer adiabatic housing: $\Delta y_1 = 0.05 \text{ mm}$, 10 layers, 1.20 growth rate.

Using the high-temperature glycerol properties, the viscous length scale near the interface is estimated at $\ell^* \approx 0.021 \text{ mm}$. The applied first layer height of 0.01 mm ensures a $y^+ \approx 0.48$, comfortably satisfying the $y^+ \leq 1$ requirement to capture the steep thermal gradients governing the convective heat transfer.

Local sizing (Solid wall through-thickness). To accurately resolve the conductive conjugate heat transfer through the 316L stainless-steel tube, edge sizing was applied to the radial thickness of the solid domain. The 0.25 mm thick wall was discretized into 5 uniform elements (yielding a radial cell size of 0.05 mm). This provides sufficient spatial resolution to accurately capture the logarithmic conduction profile across the thin metal barrier.

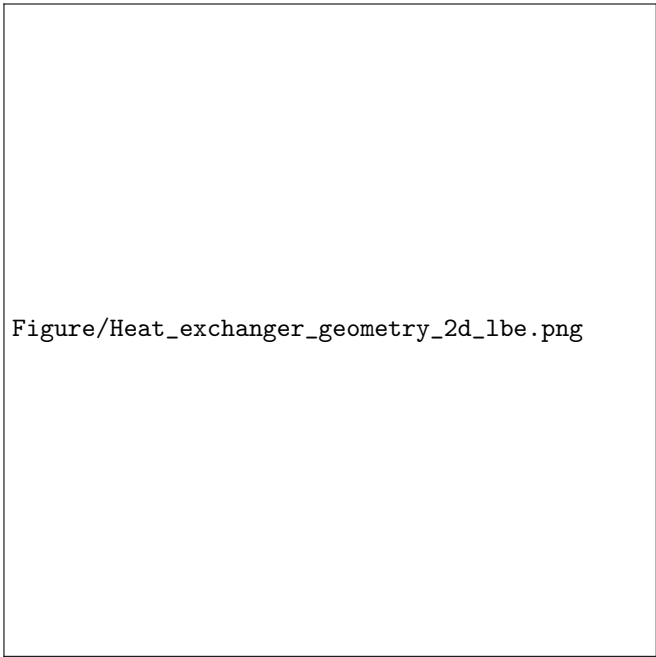
5.4 ANSYS Fluent Solver Configuration

The solver configuration builds upon the single rod setup but introduces specific modifications for the multi-fluid conjugate heat transfer problem. Since for the single rod heater it was found that the Kays correlation for Pr_t gives better results for the modeling of liquid metals, it was employed also in this case.



Figure/Heat_exchanger_geometry_2d.png

(a) General view of the heat exchanger computational grid.



Figure/Heat_exchanger_geometry_2d_lbe.png

(b) Zoomed-in view of the LBE boundary layer resolution.



Figure/Heat_exchanger_geometry_2d_glyc.png

(c) Zoomed-in view of the glycerol boundary layer resolution.

Figure 14: Overview of the structured quadrilateral mesh utilized for the straight-tube heat exchanger, highlighting the near-wall refinement required for accurate conjugate heat transfer evaluation.

Table 6: ANSYS Fluent solver settings for the straight tube heat exchanger.

Setting	Value
Solver	Pressure-based, steady-state, axisymmetric
Pressure-velocity coupling	Coupled
Energy equation	Enabled (Fluid and Solid domains)
Glycerol Viscous Model	$k-\omega$ SST with default Pr_t
LBE Viscous Model	$k-\omega$ SST with Kays correlation
Fluid-Solid Interfaces	Coupled

5.4.1 Initialization and Calculation Phases

Because the simulation involves temperature-dependent properties and highly contrasting fluids, a phased solving approach is required:

- **Initialization:** Hybrid Initialization is utilized to establish an initial approximation of the velocity and pressure fields, accelerating early convergence.
- **Patching:** The glycerol fluid zone is immediately patched to its initial experimental temperature ($T \approx 423.15$ K or 150°C) to provide a realistic starting guess for the thermal gradients and prevent unphysical property evaluation.
- **Phase 1 (Hydrodynamic):** The Energy equation is disabled. The solver runs for approximately 500 iterations to establish the velocity profiles in both fluid streams.
- **Phase 2 (Thermal Coupling):** The Energy equation is enabled with a reduced under-relaxation factor ($\alpha_T = 0.8$) to dampen temperature oscillations at the solid-fluid interfaces.

5.5 Analytical Heat Balance (ε -NTU Method)

Since the LBE outlet temperature and the total heat removed by the heat exchanger (Q_{HX}) are the primary unknowns to be verified by the CFD simulation, an analytical heat balance is performed using the ε -NTU method.

It is important to note that $Q_{\text{HX}} \neq 21$ kW at $t = 2000$ s. The facility is in a start-up transient, meaning a large fraction of the core heater power is absorbed as sensible heat by the bulk LBE inventory in the core tank rather than being immediately rejected by the heat exchanger.

1. Operating Conditions. Based on experimental data at $t = 2000$ s [Ma2009]:

- LBE Inlet: $T_{\text{LBE,in}} \approx 300^\circ\text{C}$
- Glycerol Inlet: $T_{\text{gly,in}} \approx 150^\circ\text{C}$
- LBE Mass Flowrate: $\dot{m}_{\text{LBE}} \approx 0.25$ kg s⁻¹
- Glycerol Mass Flowrate: $\dot{m}_{\text{gly}} \approx 1.4$ kg s⁻¹

2. Overall Thermal Resistance. The overall thermal resistance per unit length comprises the LBE convection, the stainless steel wall conduction, and the glycerol convection:

$$R_{\text{tot}} = \frac{1}{h_{\text{LBE}} \pi D_{\text{in}}} + \frac{\ln(D_{\text{out}}/D_{\text{in}})}{2\pi \lambda_{\text{SS}}} + \frac{1}{h_{\text{gly}} \pi D_{\text{out}}} \quad (44)$$

Using analytical correlations for turbulent LBE (Lyon-Martinelli, $h_{\text{LBE}} \approx 19550$ W m⁻² K⁻¹) and turbulent glycerol flow (Dittus-Boelter, $h_{\text{gly}} \approx 448$ W m⁻² K⁻¹), the overall heat transfer coefficient per unit length is approximately $U_L = 1/R_{\text{tot}} \approx 7.50$ W m⁻¹ K⁻¹. The convective boundary layer of the glycerol provides the vast majority of the total thermal resistance, acting as the primary bottleneck for radial heat transfer despite the convective enhancement provided by its turbulent flow regime.

3. Heat Capacity Rates and Effectiveness. The heat capacity rates are evaluated as:

$$C_{\text{LBE}} = \dot{m}_{\text{LBE}} c_{p,\text{LBE}} = 0.25 \times 146 = 36.5 \text{ W K}^{-1}, \quad (45)$$

$$C_{\text{gly}} = \dot{m}_{\text{gly}} c_{p,\text{gly}} = 1.4 \times 2427 \approx 3398 \text{ W K}^{-1}. \quad (46)$$

Because $C_{\text{LBE}} \ll C_{\text{gly}}$, the minimum heat capacity rate is heavily bottlenecked by the liquid metal ($C_{\min} = C_{\text{LBE}}$). The Number of Transfer Units (NTU) for the $L = 1$ m exchanger is:

$$\text{NTU} = \frac{U_L \cdot L}{C_{\min}} = \frac{7.50 \times 1}{36.5} \approx 0.205. \quad (47)$$

For a counter-current heat exchanger, the effectiveness is $\varepsilon = 1 - e^{-\text{NTU}} \approx 0.186$.

4. Analytical Predictions. The theoretically predicted total heat transferred is:

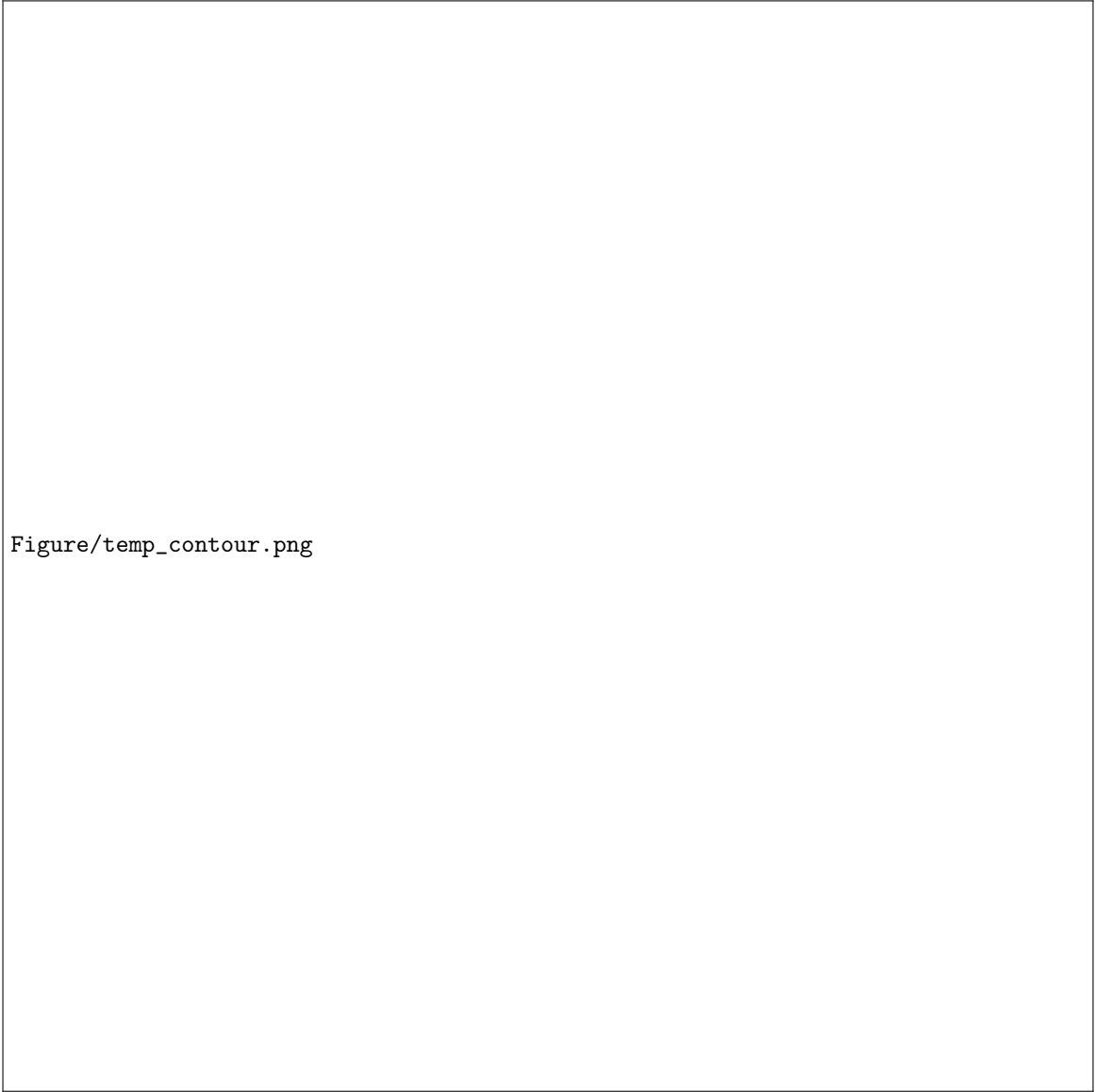
$$Q_{\text{HX}} = \varepsilon C_{\min} (T_{\text{LBE,in}} - T_{\text{gly,in}}) = 0.186 \times 36.5 \times (300 - 150) \approx 1018 \text{ W}. \quad (48)$$

The predicted steady-state outlet temperatures, which serve as the benchmark for the CFD simulation, are:

$$T_{\text{LBE,out}} = 300 - \frac{1018}{36.5} \approx 272.1^\circ\text{C}, \quad (49)$$

$$T_{\text{gly,out}} = 150 + \frac{1018}{3398} \approx 150.3^\circ\text{C}. \quad (50)$$

5.6 Results and Validation



Figure/temp_contour.png

Figure 15: Temperature contours of the straight tube heat exchanger at steady state.

The simulated thermal fields are visualized in Figure 15. To rigorously assess the conjugate heat transfer mechanisms, radial and axial temperature distributions were extracted from the CFD results, as presented in Figures 16 and 17 for the radial profiles, and Figures 18 and 19 for the axial profiles.

Figure/Fig1_Radial_LBE_Temp.png

Figure/Fig2_Radial_Glycol_Temp.png

Figure 16: Radial temperature profiles within the inner LBE flow at various axial positions.

Figure 17: Radial temperature profiles within the outer glycerol flow at various axial positions.

The radial profiles (Figures 16 and 17) highlight the fundamental difference in heat transport between the two fluids. As the LBE moves toward the outlet, its temperature decreases moving outwards radially due to heat exchange at the tube wall. Because LBE is a low-Prandtl fluid, its high thermal conductivity ensures a relatively gentle temperature gradient across the bulk. In stark contrast, the highly viscous, high-Prandtl glycerol exhibits extremely severe radial gradients. The heat entering the glycerol is largely confined to a very thin thermal boundary layer adjacent to the solid interface ($r = 6.5$ mm), leaving the bulk of the annulus completely unmixed and uniformized at the inlet temperature.

The axial profiles (Figures 18 and 19) further illustrate the counter-current arrangement. The LBE enters at $z = 1000$ mm at 300°C and steadily cools as it travels toward $z = 0$ mm. Conversely, the glycerol enters at $z = 0$ mm at 150°C . Notably, the glycerol bulk temperature remains perfectly flat along the entire 1 m length. This occurs because the heat capacity rate of the glycerol is nearly two orders of magnitude greater than that of the LBE ($C_{\text{gly}} \gg C_{\text{LBE}}$). Consequently, the glycerol acts as an infinite thermal sink, absorbing the total heat duty without experiencing a measurable rise in its bulk temperature.

Table 7: Heat exchanger validation: key output quantities comparing CFD, analytical ε -NTU predictions, and experimental data.

Quantity	CFD	ε -NTU	Experimental
LBE outlet temperature ($^\circ\text{C}$)	215.4	272.1	≈ 190
Glycerol outlet temperature ($^\circ\text{C}$)	154.5	150.3	—
Total heat transfer rate Q_{HX} (W)	2949	1018	≈ 5000

The final validation metrics are summarized in Table 7. While the CFD simulation predicts a much higher heat transfer rate (2949 W) than the idealized analytical model (1018 W), it still falls short of the true experimental heat removal (≈ 5000 W), which was taken considering a balance between the thermal power provided to the LBE in the experiment and the one that it loses in the heat exchanger. Correspondingly, the CFD LBE outlet temperature (215.4°C) is higher than the experimentally observed value ($< 200^\circ\text{C}$). This under-prediction of heat removal in the simulation can likely be attributed to the same turbulence modeling limitations identified in the single rod heater analysis. Standard RANS formulations suppress radial thermal mixing in low-Prandtl fluids. Because the numerical model does not efficiently

Figure/Fig3_Axial_LBE_Temp.png

Figure/Fig4_Axial_Glycol_Temp.png

Figure 18: Axial temperature profile of the LBE flow.

Figure 19: Axial temperature profile of the glycerol flow.

transport heat radially from the LBE core to the inner tube wall, the total energy transferred to the glycerol is artificially bottlenecked, resulting in a warmer LBE exit temperature than observed in reality.

6 Sources of Uncertainty

The discrepancies observed between the numerical predictions and the experimental data can be attributed to several overlapping sources of physical and numerical uncertainty:

- (i) **Unmodeled environmental heat losses.** In both the single rod heater and the heat exchanger simulations, the outer boundary was modeled as perfectly adiabatic. In the real TALL facility, despite external insulation, parasitic thermal losses to the ambient environment inevitably occur. This non-adiabatic reality leads to an over-prediction of the net thermal energy retained by the LBE.
- (ii) **Turbulent heat flux modeling limitations.** While the dynamic Kays Pr_t model significantly improved accuracy, the fundamental limitation lies in the inability of the standard RANS models to describe the heat transfer in liquid metal. These models inherently struggle to capture the different nature of momentum and thermal mixing in low-Prandtl liquid metals and new ad hoc models should be used.
- (iii) **Transient vs. steady-state assumptions.** The heat exchanger experimental data correspond to $t = 2000$ s from a cold-start transient. At this stage, a large fraction of the core heater power is still being absorbed as sensible heat by the bulk LBE inventory. The simulation assumes a perfect, instantaneous steady state, which inherently mismatches the transient thermal inertia of the physical loop.
- (iv) **2-D axisymmetric simplifications.** The actual TALL facility geometry includes 3-D features such as piping bends, flow distributors, grid spacers, and thermocouple penetrations. Neglecting these geometries eliminates 3-D secondary flows and mixing vortices that likely enhance the physical convective heat transfer near the inlet and outlet regions.
- (v) **Thermophysical property uncertainty.** The LBE thermal conductivity correlation carries an inherent experimental uncertainty of roughly $\pm 15\%$, which directly propagates into a proportional uncertainty in the results of the CFD simulation and of calculated thermal-hydraulic parameters.

7 Conclusions

A rigorous computational fluid dynamics (CFD) investigation of convective heat transfer in Lead-Bismuth Eutectic (LBE) was performed using 2-D axisymmetric models of the single rod heater and straight-tube heat exchanger from the

TALL facility. The numerical results were compared against experimental data, revealing fundamental limitations in standard turbulence modeling for low-Prandtl number fluids. Specifically, the study demonstrates that standard RANS models struggle to accurately represent liquid metal heat transfer due to the highly decoupled nature of the viscous and thermal boundary layers. Imposing a globally constant turbulent Prandtl number ($Pr_t \approx 1.8$) artificially suppresses radial thermal diffusion, resulting in excessive heat accumulation in the near-wall region and steep, unphysical radial temperature gradients.

While the sublayer-resolving capability of the $k-\omega$ SST model (configured with a targeted $y^+ \leq 1$) provided a more rigorous treatment of the near-wall momentum transport than the standard $k-\varepsilon$ formulation, both models ultimately struggled with the particular physics of the liquid metal. To address this, implementing the dynamic Kays correlation via a User-Defined Function (UDF) proved to be a highly effective numerical intervention. By dynamically calculating a local Pr_t based on the turbulent Péclet number, the Kays model successfully enhanced the effective thermal diffusivity in the boundary layer transition region. This localized treatment mitigated the artificial wall heat accumulation and shifted the numerical predictions significantly closer to the true experimental temperature distributions.

The conjugate heat transfer analysis of the straight-tube heat exchanger further highlighted the complex thermal mechanics of multi-fluid systems. The counter-current glycerol stream acted essentially as an infinite thermal sink due to its massive heat capacity rate ($C_{gly} \gg C_{LBE}$). However, despite operating in a transitional and turbulent flow regime, the high Prandtl number of the glycerol ($Pr \approx 45$) confined the radial heat exchange to an extremely thin convective boundary layer. This physical constraint established the secondary fluid as the primary thermal resistance and the main bottleneck for heat removal in the system.

Finally, to fully close the gap between numerical predictions and physical reality in future studies, two major modeling advancements are necessary. First, new simulations must prioritize a more physically accurate representation of the unique heat transfer mechanisms inherent to low-Prandtl number fluids. Standard models fundamentally fall short; accurately capturing the highly decoupled thermal phenomena of LBE requires implementing advanced turbulence formulations explicitly designed for liquid metals. Second, the numerical setup itself must be modeled to better reflect the real-world operational facility. The consistent over-prediction of the LBE axial temperature rise across both simulated components indicates that parasitic environmental heat losses play a non-negligible role. Replacing the idealized adiabatic boundary conditions with realistic external convective heat losses will be essential to unmask the true thermal efficiency of the experiment and achieve strict numerical validation.

A ANSYS Fluent UDF Source Code

The following C-language script contains the `DEFINE_PROPERTY` macros used to define the temperature-dependent thermophysical properties of the Lead-Bismuth Eutectic (LBE) fluid within ANSYS Fluent.

```
#include "udf.h"

DEFINE_PROPERTY(lbe_density, c, t)
{
    real T = C_T(c, t);
    return 11065.0 - 1.293 * T;          /* [kg/m^3] */
}

DEFINE_PROPERTY(lbe_viscosity, c, t)
{
    real T = C_T(c, t);
    return 4.94e-4 * exp(754.1 / T);    /* [Pa.s]   */
}

DEFINE_PROPERTY(lbe_thermal_conductivity, c, t)
{
    real T = C_T(c, t);
    return 3.284 + 1.617e-2*T - 2.305e-6*T*T; /* [W/(m.K)] */
}

DEFINE_PROPERTY(lbe_specific_heat, c, t)
{
    real T = C_T(c, t);
    return 164.8 - 3.94e-2*T + 1.25e-5*T*T
           - 4.56e5/(T*T);              /* [J/(kg.K)] */
}
```